

Generation Interconnection Request System Impact Study Report

For Queue Project Q21-1666

Roanoke Rapids 138 kV
March 29, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1666. The Interconnection Customer proposes a 43.8 MW solar photovoltaic facility interconnecting at the Roanoke Rapids 138 kV substation in the North Carolina area. The request was received on February 10, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 371 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$140,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: N Tower Tap to Cedar Falls Wi 138 kV Line
Contingency: North Tower Tap to Haggin 345 kV
Loading: 112.5% (Rating: 304 MVA)
DFAX: 6.4%
Pre-project loading: 111.6%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
N Tower Tap to Cedar Falls Wi 138 kV L	Rebuild line with higher-capacity conductor	\$7,076,000	2% (\$140,000)
	Total allocated to project		\$140,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.